

BiteFixes

Software Architecture Specification v1.0

Document Type: Software Architecture Document

Project: BiteFixes Intelligent Support Platform

Architecture Version: 1.0

Status: Technical Specification

Architecture Style: Modular SaaS + AI Orchestration Architecture

1. Executive Summary

BiteFixes is a scalable intelligent technical support platform designed to provide automated and human-assisted IT support through multiple communication channels.

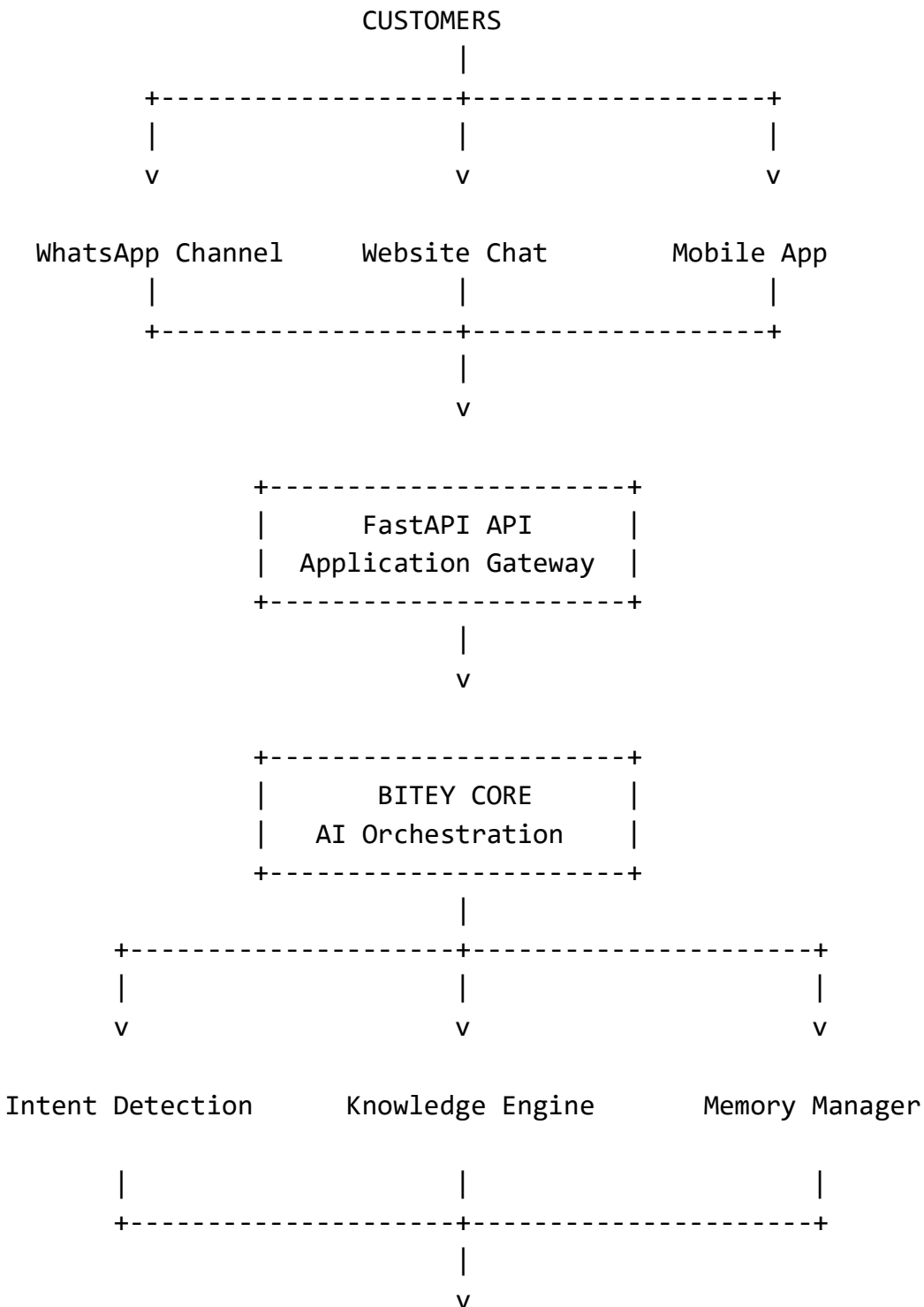
The platform uses **Bitey** as the central AI orchestration engine responsible for:

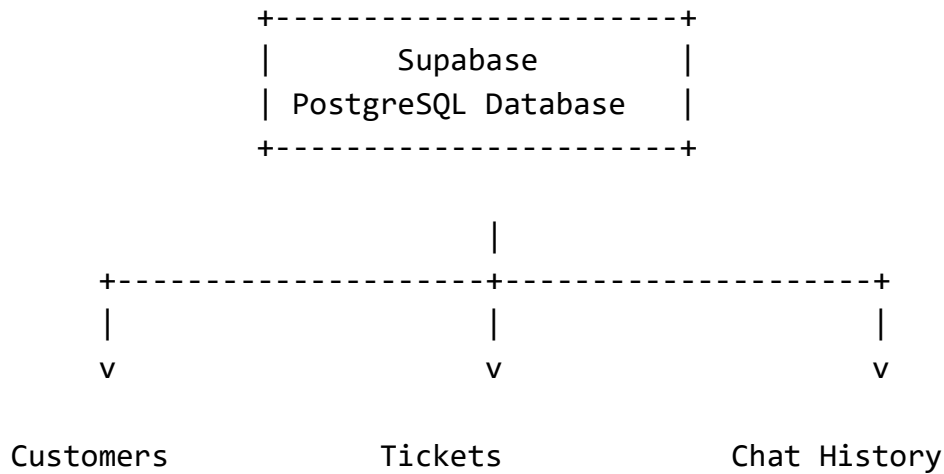
- Understanding customer requests.
- Detecting service intentions.
- Searching technical knowledge.
- Maintaining conversation memory.
- Creating and managing support tickets.
- Coordinating responses between different communication channels.

The architecture follows a modular SaaS approach where each component can evolve independently while maintaining integration through APIs.

2. High-Level Architecture Overview

System Architecture Diagram





3. Core Components

3.1 Bitey AI Core

Bitey is the intelligence layer of the platform.

Responsibilities:

Conversation Understanding

Receives messages from:

- WhatsApp
- Website chat
- Mobile application

Example:

Customer:

"Meu notebook está lento"

Processing:

Message

|

v

Normalization

|

v

Intent Detection

|

v

Knowledge Search

|

v

Response Generation

|

v

Ticket Creation (if required)

3.2 Intent Detection Engine

Purpose:

Identify customer needs automatically.

Example:

Input:

"Meu computador está muito devagar"

Detection:

Intent:

upgrade_hardware

Service:

SSD Upgrade / RAM Upgrade

Confidence:

High

Database source:

sinonimos_ia

Example:

User phrase	Detected intent
notebook lento	upgrade_hardware
tela quebrada	screen_repair
vírus	malware_removal
câmera sem imagem	cctv_problem

3.3 Knowledge Base Engine

Technical knowledge repository.

Database:

base_conhecimento

Stores:

- Troubleshooting guides
- Service information
- FAQ
- Solutions
- Procedures

Example:

Question:

"Meu notebook está lento"

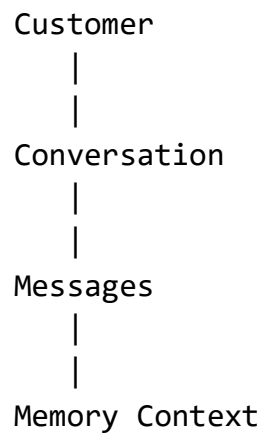
Answer:

"Podemos realizar upgrade com SSD
e aumento de memória RAM."

3.4 Conversational Memory

Responsible for remembering customer history.

Architecture:



Tables:

clientes

conversaciones

historial_chats

4. Backend Architecture

Technology Stack:

Backend:

Python
FastAPI

Database:

Supabase PostgreSQL

API:

REST

Deployment:

Render

Frontend:

WordPress
Flutter Mobile App

Backend Internal Structure

app/

├─ main.py

├─ routers/

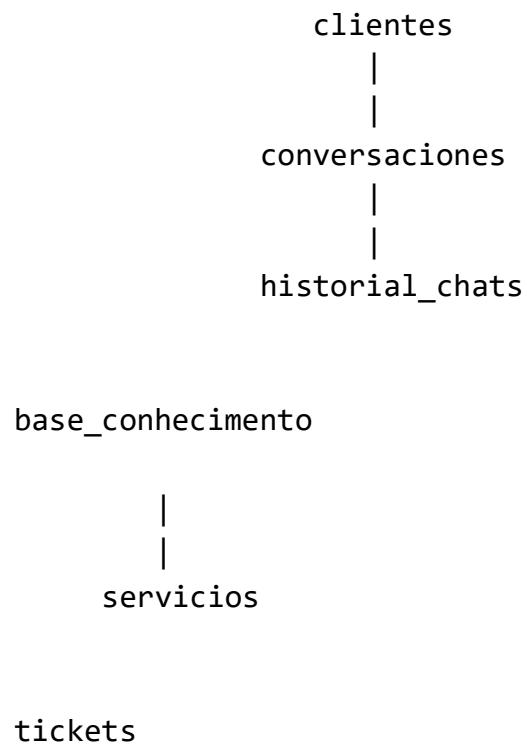
| ├─ chat.py
| ├─ clientes.py
| └─ tickets.py

└─ services/

```
|   ├── bitey.py
|   ├── intencion_service.py
|   ├── conocimiento_service.py
|   ├── historial_service.py
|   └── cliente_service.py
|
|── models/
|
|── database/
|
|   └── supabase.py
```

5. Database Architecture

Supabase Data Model



|
|
clientes

sinonimos_ia

|
|
intent_detection

Main Tables

clientes

Stores:

- Customer identity
- WhatsApp number
- Contact information

conversaciones

Stores:

- Conversation sessions
- Channel source
- Status

historial_chats

Stores:

- User messages
- Bot responses
- Timestamps

tickets

Stores:

- Technical requests
- Status
- Assigned service

Example:

BF-2026-000007

Status:

Aberto

Service:

Notebook Performance

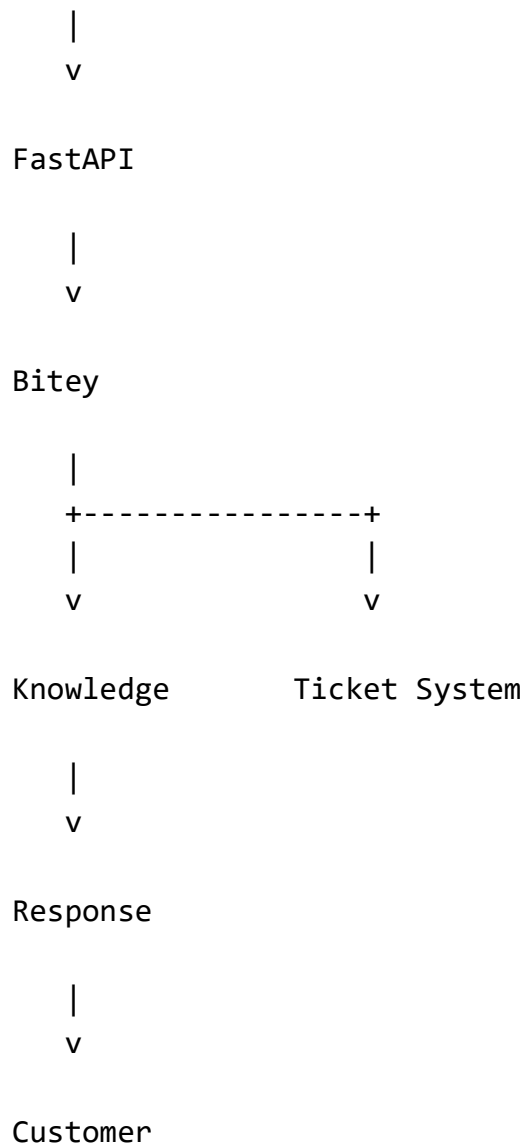
6. Communication Flow

Customer Support Flow

Customer

|
v

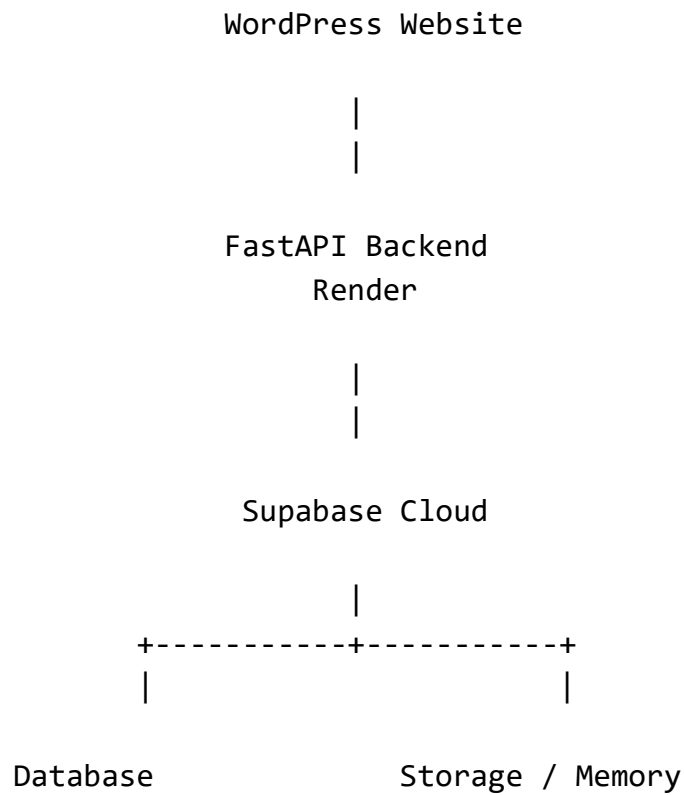
WhatsApp/Web/App



7. Deployment Architecture

INTERNET

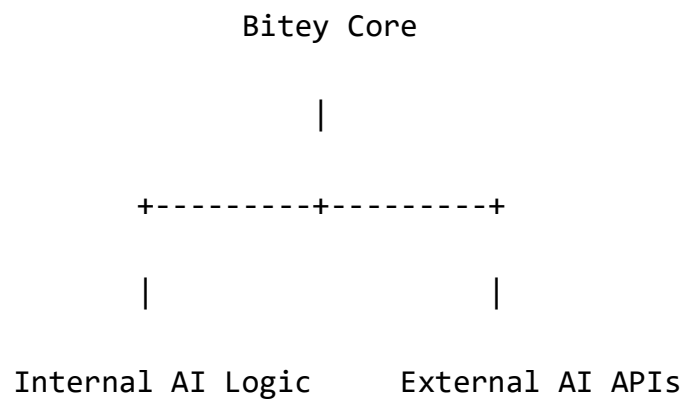


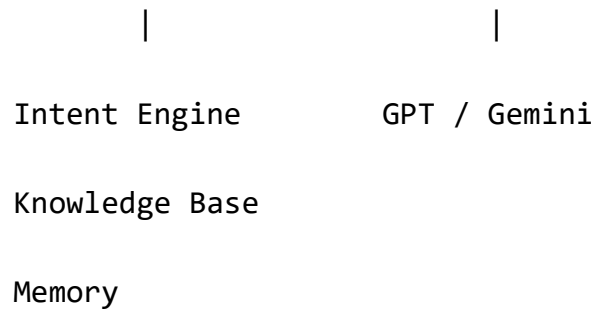


8. External AI Integration Strategy

External AI providers are considered enhancement layers.

Architecture:





Principle:

Bitey remains the owner of business logic and customer intelligence.

9. Security Architecture

Security principles:

- API authentication.
- Environment variables for secrets.
- Database access control.
- Customer data isolation.
- Logging and monitoring.

10. SaaS Scalability Roadmap

Future evolution:

Phase 1

Current:

- ✓ FastAPI
- ✓ Supabase

- ✓ Bitey Core
- ✓ Knowledge Base
- ✓ Ticket Automation

Phase 2

Expansion:

- WhatsApp Cloud API
- Flutter application
- Customer portal
- Technician dashboard

Phase 3

Enterprise SaaS:

- Multi-tenant architecture
- Multiple companies
- Billing system
- Analytics dashboard
- AI optimization

11. Architecture Principles

BiteFixes follows:

Modular Design

Each service has independent responsibility.

API First

All channels communicate through APIs.

AI Orchestration

Bitey controls intelligence flow.

Data Ownership

Customer memory remains inside BiteFixes infrastructure.

Multilingual Ready

Supported languages:

- Spanish
- Portuguese
- English

Document Status

BiteFixes Software Architecture Specification v1.0

Prepared for:

BiteFixes Intelligent Technical Support Platform

Architecture Owner:

BiteFixes Development Team

BITEFIXES

Software Architecture Specification

Version 1.0

Intelligent Technical Support SaaS Platform

Architecture Type

Modular AI-Powered SaaS Architecture

Core Engine

Bitey AI Support Orchestrator

Backend

Python + FastAPI

Database

Supabase PostgreSQL

Channels

- WhatsApp
- Website Chat
- Mobile Application

Deployment

Cloud Native Architecture

Document Information

Field	Value
Project	BiteFixes
Version	1.0
Architecture	SaaS + AI Orchestration
Backend	FastAPI
Database	Supabase
AI Core	Bitey
Languages	ES / PT / EN
Status	Active Development

TABLE OF CONTENTS

1. Executive Summary
2. Business Vision
3. System Architecture
4. Component Architecture
5. Bitey AI Architecture
6. Backend Architecture
7. Database Architecture
8. UML Component Diagram
9. Sequence Diagrams
10. Entity Relationship Model
11. Deployment Architecture

- 12. Security Architecture
- 13. Ticket Management Workflow
- 14. External AI Integration
- 15. Scalability Roadmap

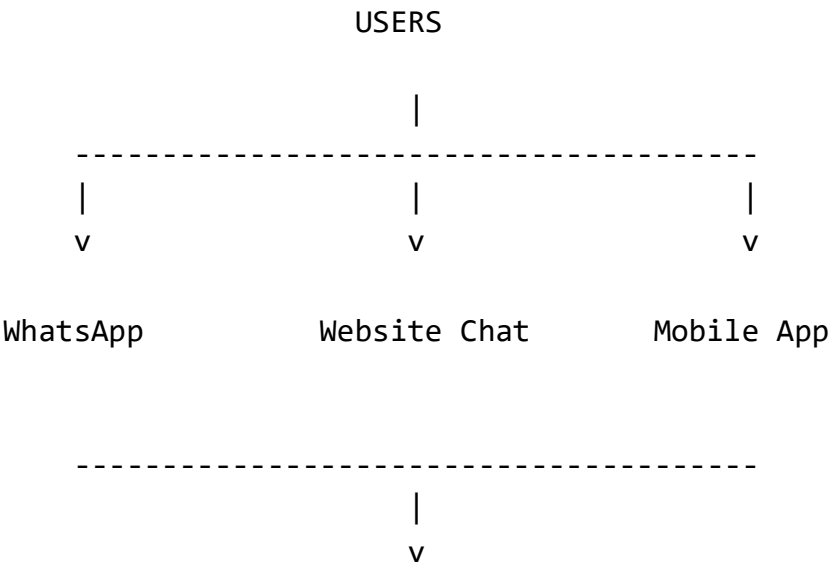
1. Business Vision

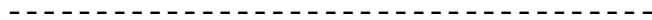
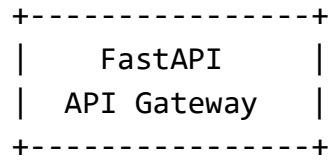
BiteFixes aims to become an intelligent technical support platform capable of automating IT assistance while maintaining human-quality service.

The platform combines:

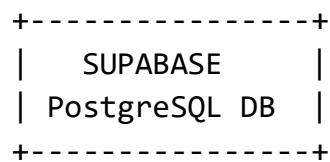
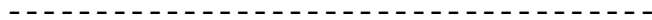
- Artificial Intelligence
- Technical Knowledge Management
- Customer Memory
- Automated Ticketing
- Multi-channel Communication

2. Complete System Architecture

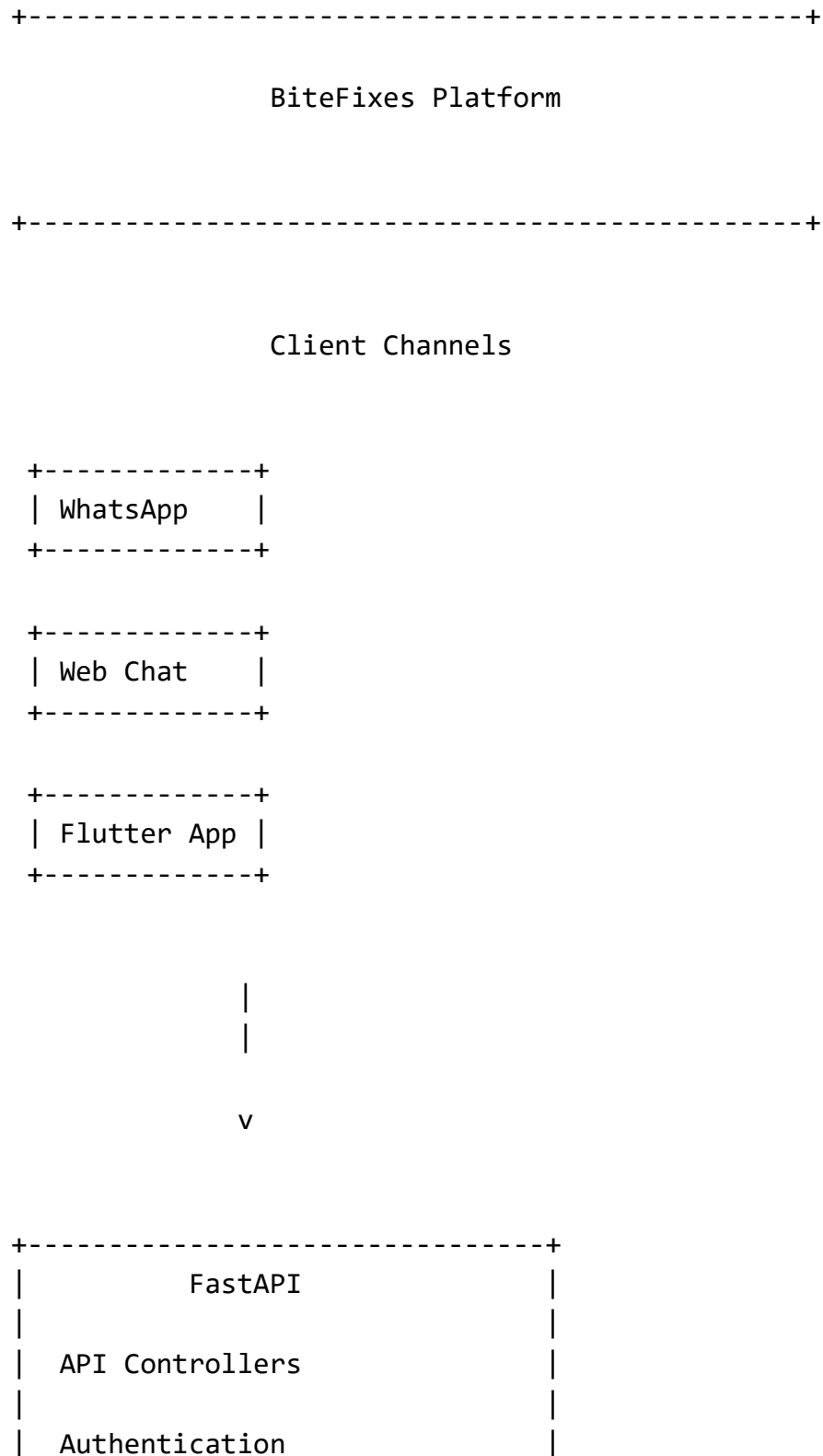


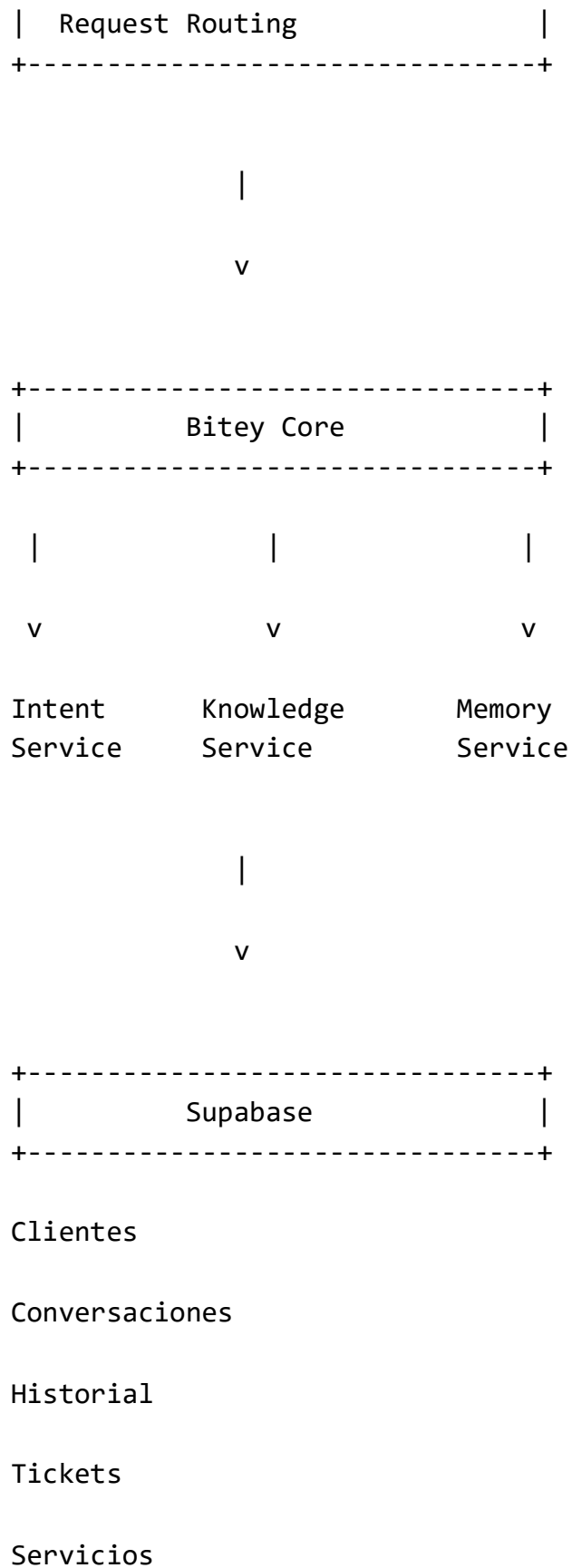


Intent Engine Knowledge Base Memory Engine



3. UML Component Diagram





Base Conhecimento

Sinonimos IA

4. Sequence Diagram

Customer Chat Flow

Customer

```
|
| "Meu notebook está lento"
|
v
```

WhatsApp/Web/App

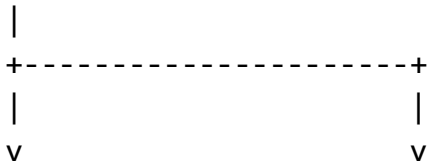
```
|
|
v
```

FastAPI Endpoint

/chat

```
|
|
v
```

Bitey Core



Intent Detection

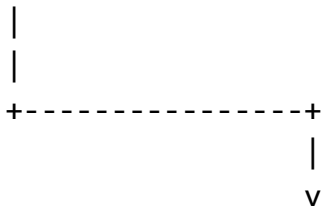
Memory Search



Knowledge Base



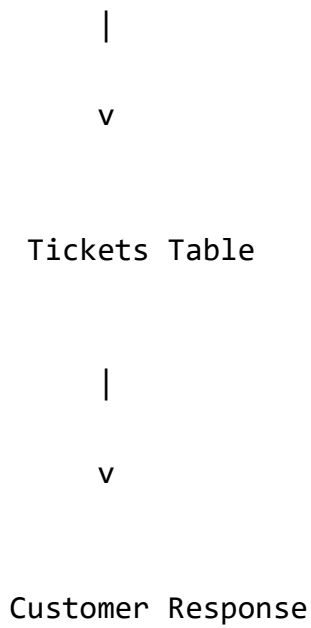
Generate Response



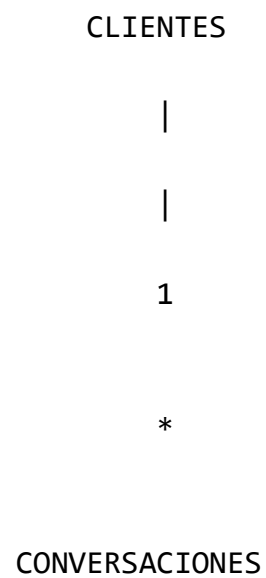
Create Ticket?



Yes ----- No



5. Database Entity Relationship Diagram



|

|

1

*

HISTORIAL_CHATS

CLIENTES

|

|

*

TICKETS

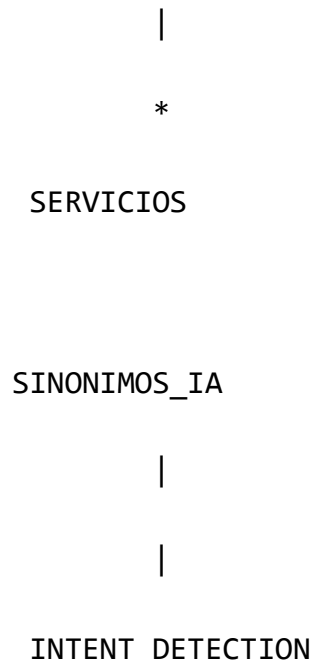
|

|

SERVICIOS

BASE_CONHECIMENTO

|



6. Supabase Database Model

clientes

id
nome
telefone
email
created_at

conversaciones

id
cliente_id
canal
estado

created_at

historial_chats

id
conversacion_id
remitente
mensaje
fecha

tickets

id
codigo_ticket
cliente_id
servicio_id
status
prioridad

Example:

BF-2026-000007

Status:

Aberto

7. Ticket Automation Workflow

Customer Problem

|

v

Intent Detection

|

v

Service Identification

|

v

Ticket Creation

|

v

Technician Assignment

|

v

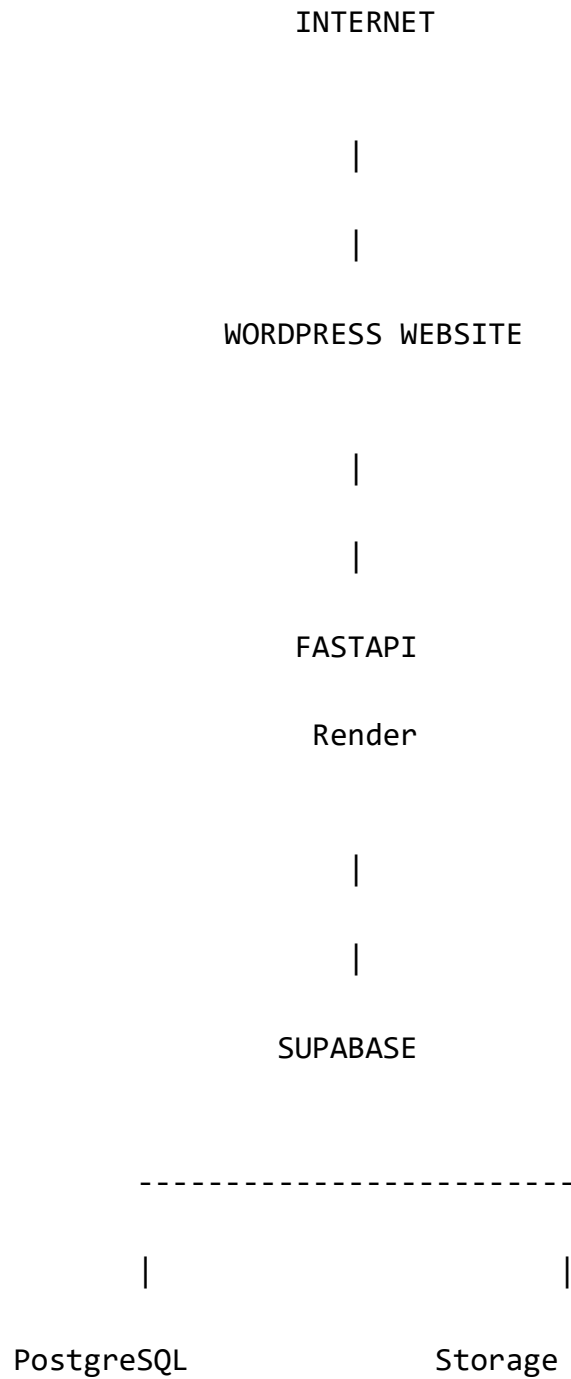
Resolution

|

v

Customer Notification

8. Deployment Architecture



9. Security Architecture

Application Security

User

|

HTTPS

|

FastAPI Security Layer

|

Authentication

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Authorization

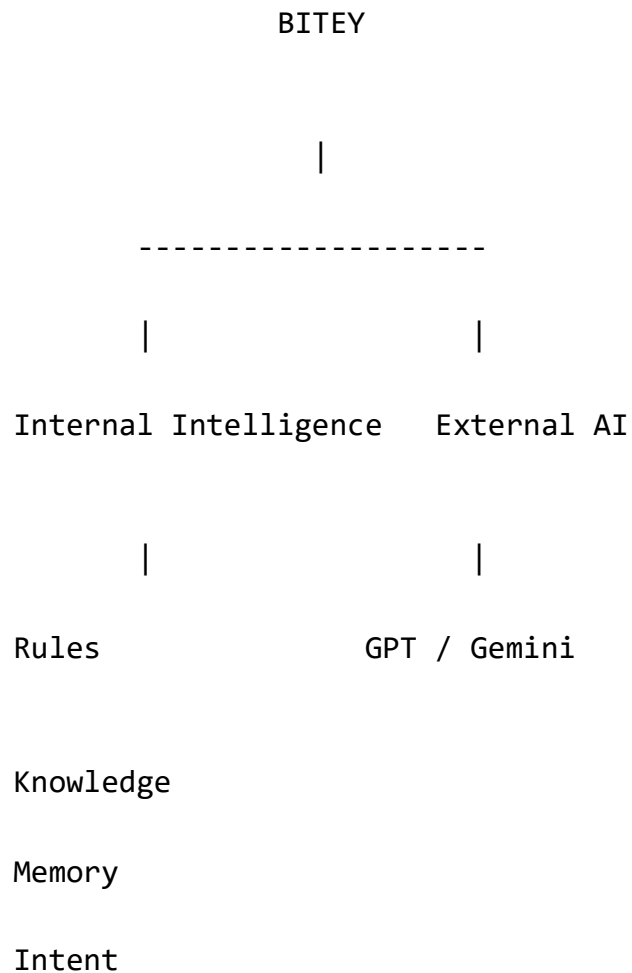
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Database Policies

Security controls:

- HTTPS
- Environment variables
- API keys protection
- Database permissions
- Input validation
- Logging

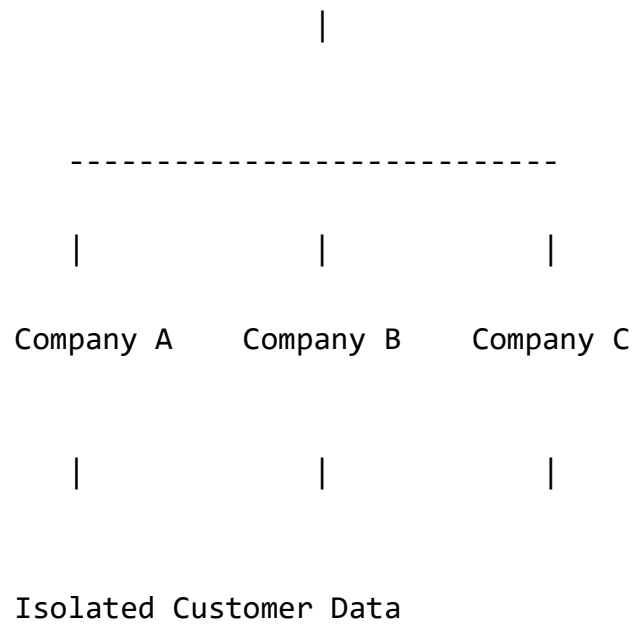
10. AI Architecture Strategy



External AI improves capability but does not replace Bitey.

11. SaaS Multi-Tenant Future Architecture

Bitey Platform



12. Technology Stack

Layer	Technology
Backend	Python FastAPI
Database	Supabase PostgreSQL
AI Core	Bitey
Frontend	WordPress
Mobile	Flutter
Hosting	Render
Authentication	API Security
Storage	Cloud Storage

13. Development Roadmap

Phase 1 - Foundation

Completed / Current:

- ✓ FastAPI
- ✓ Supabase connection
- ✓ Customer management
- ✓ Knowledge base
- ✓ Intent detection
- ✓ Chat memory

Phase 2 - Automation

Next:

- WhatsApp Cloud API
- Automatic ticket workflows
- Technician dashboard
- Notifications

Phase 3 - SaaS Platform

Future:

- Multi-company support
- Subscription plans
- Analytics
- AI optimization
- Mobile application

14. Final Architecture Statement

BiteFixes is designed as a scalable AI-powered technical support ecosystem where:

Channels communicate with FastAPI.

FastAPI communicates with Bitey.

Bitey manages intelligence, memory and workflows.

Supabase stores business knowledge and customer history.

This architecture allows BiteFixes to evolve from a local technical support business into a complete SaaS support platform.

END OF DOCUMENT

BiteFixes Software Architecture Specification v1.0